Exam: Error Calculation & Measurement Uncertainty

Practice Exam with Solutions | Basic Level

1. [Application] (4 Pts.) Reaction times: 220,245,235,210,240 ms. Mean, SD, SE, 95% CI. n for SE<5ms?

2. [Application] (4 Pts.) Density rho=m/V, m=100+/-1g, V=50+/-2cm^3. rho with error. Relative error?

3. [Open question] (3 Pts.) Gaussian: Y=X1+X2, sigma1=0.5, sigma2=1.2. sigma_Y? Y=X1*X2, M1=10, M2=5?

4. [Open question] (2 Pts.) Systematic vs random errors? Examples?

5. [Open question] (2 Pts.) When Monte Carlo instead of Gaussian propagation?

SOLUTIONS

1. Mean=230, SD=14.79, SE=6.61. CI=230+/-2.776*6.61=[211.6,248.4]. n=9.

2. rho=2.00+/-0.08 g/cm^3. Relative=4.1%.

3. sigma_Y=1.30. Y=50+/-12.26.

4. Systematic: consistent direction (calibration). Random: scatter around true value (reading error).

5. Nonlinear functions, non-normal inputs, correlated inputs.